1. Title

Autonomous IoT-Based Emergency Detection and Response System for Disabled Individuals

---

2. Idea

The project aims to develop a smart IoT system that continuously monitors a disabled individual's health and physical condition and automatically detects emergencies such as falls, abnormal vital signs, or immobility. Upon detecting a critical situation, the system will autonomously trigger alerts to caregivers or emergency services without requiring user intervention.

This system is particularly useful for individuals living alone who may be unable to call for help during emergencies.

---

3. Features

- Real-time health monitoring (heart rate, SpO_2, temperature)
- Fall detection using motion sensors
- Inactivity detection
- Location tracking via GPS
- Automatic emergency alert (SMS/Call/App notification)
- Multi-sensor decision mechanism
- No manual input required
- Caregiver monitoring dashboard
- Local alarm/buzzer activation

---

4. Methodology

1. Collect physiological and motion data using wearable sensors.
2. Process sensor data using a microcontroller (ESP32).
3. Implement emergency detection logic combining vital signs and motion patterns.
4. Identify abnormal conditions such as fall + immobility or critical health parameters.
5. Transmit emergency data to cloud/server via Wi-Fi or GSM.
6. Automatically notify caregivers with patient status and location.
7. Provide real-time monitoring through a web/mobile interface.
8. Test system performance for accuracy and reliability.

---

5. Components

Hardware Components

- ESP32 Microcontroller
- MAX30102 Heart Rate & SpO_2 Sensor
- MPU6050 Accelerometer & Gyroscope
- Temperature Sensor (LM35/DS18B20)
- GPS Module (NEO-6M)
- GSM Module (SIM800L) or Wi-Fi
- Buzzer / Alarm
- Rechargeable Battery

Software Components

- Arduino IDE
- IoT Cloud Platform (ThingSpeak/Firebase)
- Web or Mobile Application

---

6. Block Diagram